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Modular extracellular sensor architecture for regulating genes

Abstract

Disclosed are systems, components of systems, and methods for sensing extracellular ligands and/or modulating expression of an endogenous or exogenous gene in a cell. In some embodiments, the disclosed systems and methods comprise or utilize first and second exogenous extracellular sensors and/or nucleic acid sequences encoding the first and second exogenous extracellular sensors, wherein the first exogenous extracellular sensor comprises: a) a ligand binding domain; b) a transmembrane domain; c) a protease cleavage site; and d) a functional domain comprising an RNA-binding subdomain fused to a transcription regulatory subdomain; and the second exogenous extracellular sensor comprises: e) a ligand binding domain, f) a transmembrane domain, and g) a protease domain that cleaves the protease cleavage site of the first exogenous extracellular receptor.

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Background/Summary

CROSS-REFERENCE TO RELATED PATENT APPLICATIONS (1) This application claims the benefit of priority under 35 U.S.C. § 119(e) to U.S. Provisional Application No. 62/411,895, filed on Oct. 24, 2016, the content of which is incorporated herein by reference in its entirety.

BACKGROUND

(1) The present invention provides modular extracellular sensors, nucleic acids encoding such sensors, cells expressing such sensors, and methods of employing such sensors and cells for sensing extracellular ligands and/or modulating expression of a gene in a cell when the ligand for the sensors is present.

SUMMARY

(2) Disclosed are systems, compounds, compositions, and methods for sensing extracellular ligands and/or modulating expression of genes in a cell including endogenous genes and exogenous genes. In some embodiments, the disclosed systems and methods comprise or utilize: i) an exogenous extracellular sensor, and/or ii) a nucleic acid sequence encoding the exogenous extracellular sensor, wherein the exogenous extracellular sensor comprises: a) a ligand binding domain, b) a transmembrane domain, c) a protease cleavage site, and d) a functional domain comprising an RNA-binding subdomain fused to a transcription regulatory subdomain. The exogenous extracellular sensor may be referred to as a recombinant fusion protein comprising a) the ligand binding domain, b) the transmembrane domain, c) the protease cleavage site, and d) the functional domain comprising an RNA-binding subdomain fused to a transcription regulatory subdomain, wherein one or more of the domains of the fusion protein may be heterologous to one or more other domains of the fusion protein.

(3) Suitable RNA-binding subdomains of the functional domain of the exogenous extracellular sensor may include RNA-binding domains found in eukaryotic proteins and/or prokaryotic proteins. In some embodiments, the RNA-binding subdomain comprises a RNA-binding domain of a Cas9 protein. The RNA-binding subdomain of the functional domain may bind an RNA that confers DNA recognition to the functional domain and/or that targets the functional domain to a DNA target such as a transcription promoter. Suitable RNAs bound by the RNA-binding subdomain of the functional domain may include small guide RNAs (sgRNAs) that may be utilized to target the functional domain to a DNA target.

(4) In some embodiments, the disclosed systems and methods comprise or utilize: i) first and second exogenous extracellular sensors, and/or ii) one or more nucleic acid sequences encoding the first and second exogenous extracellular sensors, wherein the first exogenous extracellular sensor comprises: a) a ligand binding domain, b) a transmembrane domain, c) a protease cleavage site, and d) a functional domain comprising an RNA-binding subdomain fused to a transcription regulatory

subdomain; and wherein the second exogenous extracellular sensor comprises: e) a ligand binding domain, f) a transmembrane domain, and g) a protease domain. In systems where the first and second exogenous extracellular sensors are present in a cell as membrane proteins, the protease domain of the second exogenous extracellular sensor may recognize and cleave the protease cleavage site of the first exogenous extracellular sensor, for example, when the first exogenous extracellular sensor and the second exogenous extracellular sensor bind to an extracellular ligand via the ligand binding domain of the first exogenous extracellular sensor and the ligand binding domain of the second exogenous extracellular sensor, respectively. In some embodiments, the first exogenous extracellular sensor and the second exogenous extracellular sensor bind to the same ligand, and when the first exogenous extracellular sensor and the second exogenous extracellular sensor bind to the same ligand, the protease domain of the second exogenous extracellular sensor is positioned near the protease cleavage site of the first exogenous extracellular sensor to facilitate cleavage of the protease cleavage site of the first exogenous extracellular sensor by the protease domain of the second exogenous extracellular sensor. Through this cleavage, the functional domain of the first exogenous extracellular sensor comprising an RNA-binding subdomain fused to a transcription regulatory subdomain then may be released from the first exogenous extracellular sensor and may migrate to a targeted gene, for example, via a sgRNA bound by the RNA-binding subdomain of the functional domain. Thus released and targeted, the functional domain may function to modulate expression of the targeted gene.

(5) The disclosed systems and methods comprising and/or utilizing the disclosed extracellular sensors may be utilized for sensing extracellular ligands and/or modulating expression of an endogenous or exogenous gene. In some embodiments, the disclosed systems and methods further may comprise or utilize an RNA or a vector that expresses an RNA that targets the functional domain of an extracellular sensor to a target gene. In the disclosed systems and methods, optionally the RNA-binding subdomain is a Cas9 RNA-binding domain and the RNA is a small guide RNA (sgRNA) that is bound by the RNA-binding subdomain of the functional domain and that targets the functional domain to a target gene whose transcription is modulated by the transcription regulatory subdomain of the functional domain.

Description

BRIEF DESCRIPTION OF THE FIGURES

(1) FIG. 1A-1E. VEGF-MESA receptor development. FIG. 1A Upper: schematic of the VEGF-MESA mechanism, by which ligand binding-induced dimerization results in trans-cleavage and release of a previously sequestered transcription factor. Lower: crystal structures of two Fabs indicating domains within each Fab binding to each of two epitopes on VEGF.sup.37. Table inset: reported equilibrium dissociation constants for Fabs binding to human and mouse VEGF.sup.22. FIG. 1B Cell-surface expression of representative VEGF-MESA constructs was evaluated by flow cytometry in cells transfected with vectors expressing VEGF-MESA versus vector-only transfected controls. Variants shown here differ in the lengths (in amino acids) of the non-structured extracellular linkers (ECL). Gray area of graphs represents vector-only transfected controls. This experiment is representative of two independent experiments. FIG. 1C VEGF-binding to VEGF-MESA was quantified using flow cytometry. Bar heights correspond to mean fluorescence intensities of samples comprising at least 5000 transfected cells, and each error bar represents one standard deviation. This experiment is representative of two independent experiments. FIG. 1D VEGF-inducible signaling by V1-MESA was quantified by flow cytometry. "Reporter activity" was calculated by quantifying mean fluorescence intensity (MFI) of YFP in transfected cells, then normalizing this value to that calculated for the internal control (cells transfected with reporter-only, the value of which was set to one). Numbers above the bars represents fold induction,

calculated by dividing the reporter signaling in the presence of VEGF by the reporter signaling in the absence of VEGF. Experiments were conducted in biological triplicate, each experiment is representative of at least three independent biological experiments, and error bars represent one standard deviation. (* $p \leq 0.05$, *** $p \leq 0.001$). FIG. 1E VEGF-inducible signaling by V2-MESA was quantified by flow cytometry. “Reporter activity” was calculated by quantifying mean fluorescence intensity (MFI) of YFP in transfected cells, then normalizing this value to that calculated for the internal control (cells transfected with reporter-only, the value of which was set to one). The boxed area highlights constructs exhibiting significant VEGF-inducible signaling. Numbers above the bars represents fold induction, calculated by dividing the reporter signaling in the presence of VEGF by the reporter signaling in the absence of VEGF. Experiments were conducted in biological triplicate, each experiment is representative of at least three independent biological experiments, and error bars represent one standard deviation. (* $p \leq 0.05$, *** $p \leq 0.001$)

(2) FIG. 2A, FIG. 2B FIG. 2C, FIG. 2D, FIG. 2E, and FIG. 2F. VEGF-MESA receptor implementation. FIG. 2A This cartoon illustrates potential MESA complexes expected to form upon VEGF binding-induced dimerization. FIG. 2B This cartoon illustrates the proposed mechanism by which ‘dead’ TC complexes may form due to transient receptor contact during trafficking or on the cell surface. FIG. 2C, FIG. 2D, and FIG. 2E Both overall expression levels and ratios of the two VEGF-MESA chains was varied by dosing in different amounts of plasmid encoding each of the two MESA chains (TC and PC), as indicated in both the table (plasmid ratio) and ramps (plasmid masses) placed under each panel. Boxes highlight the regimes in which VEGF-inducible receptor signaling was observed. Numbers above the bars represents fold induction and data were analyzed as described in FIG. 1. Fig. F Dose response of VEGF-MESA. Based upon results shown in FIG. 2C, FIG. 2D, and FIG. 2E, VEGF-MESA chains were expressed at a TC:PC ratio of 24 (3 μ g TC plasmid and 0.125 μ g PC plasmid per well). Horizontal bars highlight the 0 ng/mL VEGF sample, to which each test sample was compared, with the thickness indicating one standard deviation. Sigmoidal lines were added to guide the eye. Data were analyzed as described in FIG. 1. (* $p \leq 0.05$, ** $p \leq 0.01$).

(3) FIG. 3A, FIG. 3B, FIG. 3C, FIG. 3D, FIG. 3E, FIG. 3F, FIG. 3G, FIG. 3H, and FIG. 3I. Rewiring of cellular input-output using MESA. FIG. 3A Identification of sgRNAs that confer dCas9-TF-mediated induction of IL-2 expression. sgRNAs were screened individually and in combination and IL-2 mRNA expression was measured by qPCR. IL-2 mRNA copy number was normalized to GAPDH mRNA copy number, and this ratio was normalized to that calculated for cells transfected with only dCas9-TF (black bar). Labeled squares in the table indicate sgRNA that, individually, confer substantial IL-2 expression and were selected for further analysis. mRNA concentrations were evaluated in technical triplicate, and results are representative of three independent biological experiments. FIG. 3B and FIG. 3C Evaluation of SAM system^{sup.34} for enhancing dCas9-TF-induced IL-2 gene expression. Data were analyzed as in FIG. 3A. FIG. 3D and FIG. 3E These cartoons summarize the proposed scheme for rewiring a cell to generate a non-natural function, whereby exposure to a canonically immunosuppressive cue (VEGF) results in the production of a canonically immunostimulatory factor in response (IL-2). FIG. 3F-I IL-2 secretion driven by soluble dCas9-TF was evaluated in HEK 293FT cells (FIG. 3F) and Jurkat T cells (FIG. 3G) and quantified by ELISA. VEGF-inducible secretion of IL-2 by cells expressing VEGF-MESA-dCas9-TF at a TC:PC ratio of 10 and sgRNA 7 was evaluated in HEK 293FT cells (FIG. 3H) and Jurkat T cells (FIG. 3I). Each sample was quantified in technical triplicate and data shown are representative of three independent biological experiments (see FIG. 11). Numbers above the bars represents fold induction, as described in FIG. 1. Error bars represent one standard deviation. (* $p \leq 0.05$, ** $p \leq 0.01$).

(4) FIG. 4A, FIG. 4B, FIG. 4C, FIG. 4D, and FIG. 4E. Surface expression and VEGF binding of VEGF-MESAs. FIG. 4A, FIG. 4B, and FIG. 4C Cell-surface expression of VEGF-MESA was evaluated by flow cytometry (See Methods and FIG. 1 for measurement details). Variants shown

here differ in the lengths (in amino acids) of the non-structured extracellular linkers (ECL). Gray areas of graphs represent vector-only transfected controls. This experiment is representative of two independent experiments. FIG. 4D and FIG. 4E VEGF-binding to all reported VEGF-MESA clones was quantified using flow cytometry (see Methods and FIG. 1 for measurement details). Bar heights corresponds to sample mean fluorescence intensities and each error bar represents one standard deviation.

(5) FIG. 5A-FIG. 5C. Impact of intracellular linker on cell surface expression. Cell surface staining of various TC and PC constructs was quantified using immunohistochemistry and flow cytometry (see Methods and FIG. 1 for measurement details). Gray areas of graphs represent vector only transfected cell controls. FIG. 5A The two left panels indicate the original VEGF-MESA constructs, where cell surface expression for the TC was much greater than for PC, when the PC lacked an intracellular linker (ICL) between transmembrane and protease domains. PC constructs were evaluated with a 3 amino acid flexible linker between the transmembrane domain and protease (right). FIG. 5B and FIG. 5C Sequential addition of amino acids derived from the cleavage sequence. The box in FIG. 5B indicates that no further improvements in surface expression were observed for linkers beyond this length; this PC design was used for all subsequent experiments in this study.

(6) FIG. 6A, FIG. 6B, FIG. 6C, FIG. 6D, FIG. 6E, FIG. 6F, and FIG. 6G. Evaluation of VEGF-MESA with G cleavage sequence on the TC. FIG. 6A, FIG. 6B, and FIG. 6C Cell-surface staining of VEGF-MESA was quantified using immunohistochemistry and flow cytometry (see Methods and FIG. 1 for measurement details). Gray areas of graphs represent vector only transfected cell controls. FIG. 6D and FIG. 6E VEGF-binding to all reported VEGF-MESA clones was quantified using flow cytometry (see Methods). Bar heights corresponds to sample mean fluorescence intensities and each error bar represents one standard deviation. FIG. 6F and FIG. 6G VEGF-induced reporter activation for VEGF-MESA with 'G' cleavage sequence on the TC was quantified using flow cytometry. See FIG. 1 for measurement details.

(7) FIG. 7A, FIG. 7B, FIG. 7C, FIG. 7D, FIG. 7E, and FIG. 7F. VEGF-MESA surface expression levels as a function of plasmid dose. Cell surface expression of VEGF-MESA was evaluated for the range of plasmid doses evaluated in FIG. 2 for the TC (FIG. 7A, FIG. 7B, FIG. 7C) and PC (FIG. 7D, FIG. 7E, FIG. 7F) using flow cytometry (See Methods and FIG. 1 for measurement details). This experiment was conducted in biological duplicate with the histograms (FIG. 7A, FIG. 7B, FIG. 7D) showing a representative example and the line graphs FIG. 7C and FIG. 7F quantifying the normalized average expression level observed across experiments. All samples were normalized to a vector only control, and to compare between the biological replicates, the highest value for each experiment (3 g plasmid for V1-MESA TC) was set to unity. Error bars represent one standard deviation.

(8) FIG. 8A, FIG. 8B, FIG. 8C, and FIG. 8D. Tuning MESA expression levels by varying protein translation initiation rate. Protein translation initiation rate was varied by utilizing a range of synthetic upstream open reading frames (uORF). The three letter codes above denote the last three nucleotides of the Kozak sequence utilized for that particular sample. Samples with a '/' have an additional uORF added in the 5'UTR—i.e., CCC/ACC has an ACC Kozak sequence directly before the start codon for VEGF-MESA as well as one uORF in the 5' UTR, in which the last three nucleotides of the Kozak sequence are CCC. The ACC Kozak sequence is the consensus sequence that was used for the constructs in FIGS. 1, 2, and 3. FIG. 8A, FIG. 8B, FIG. 8C Cell-surface expression of uORF-containing V1-MESA constructs was evaluated utilizing flow cytometry. This experiment was conducted in biological duplicate with the histograms FIG. 8A and FIG. 8B showing a representative example and the line graph FIG. 8C quantifying the normalized average expression level observed across experiments. All samples were normalized to a vector only control, and to compare between the biological replicates, the highest value for each experiment (CGA for V1-MESA PC) was set to unity. FIG. 8D VEGF-inducible signaling was quantified by

flow cytometry for uORF-containing V1-MESA constructs. The two uORFs that conferred the highest levels of surface expression, CGA and ACC, were utilized for the TC, and these plasmids were transfected at a 1:1 ratio with one of various plasmids encoding the PC. Numbers above the bar represent fold induction, and data were analyzed as described in FIG. 1. Error bars represent one standard deviation. (* $p \leq 0.05$, ** $p \leq 0.01$).

(9) FIG. 9. Location and sequence of IL-2 sgRNAs. Schematic of locations for all IL-2 sgRNAs evaluated. Locations represent distance upstream or downstream from the transcription start site. Full sgRNA targeting sequence are listed in the table. Uppercase letters represent direct base pairing interactions with the target DNA, lowercase letters represent deviations from the target sequence. Sequences that start with a lowercase 'g' were designed in this manner to enable transcription from a U6 promoter.

(10) FIG. 10. Surface staining of VEGF-MESA with dCas9-TF. Cell-surface staining of VEGF-MESA expressed in either HEK 293FT cells or Jurkat T cells was quantified using immunohistochemistry and flow cytometry (see Methods and FIG. 1 for measurement details). Gray areas of graphs represent vector only transfected cell controls.

(11) FIG. 11A and FIG. 11B. Biological repeats for VEGF-MESA with dCas9-TF. FIG. 11A and FIG. 11B HEK 293FT cells or Jurkat T cells were transfected to transiently express VEGF-MESA with dCas9-TF and sgRNA 7, and IL-2 protein production was quantified by ELISA. TC:PC ratio was set to 10 and refers to relative amounts of TC and PC plasmids included in the transfection. Numbers above the bars represent fold induction (See FIG. 1). See FIG. 3 for measurement details. (* $p \leq 0.05$).

(12) FIG. 12A and FIG. 12B. Investigating alternative TC:PC ratios and choice of sgRNA for VEGF-MESA with dCas9-TF. FIG. 12A HEK 293FT cells were transfected to transiently express VEGFMESA with dCas9-TF and sgRNA 17, and IL-2 protein production was quantified by ELISA. TC:PC ratio was set to 10 and refers to relative amounts of TC and PC plasmids included in the transfection. Numbers above the bars represent fold induction (See FIG. 1). See FIG. 3 for measurement details. FIG. 12B HEK 293FT cells were transfected to transiently express VEGF-MESA with dCas9-TF and sgRNA 7, and IL-2 protein production was quantified by ELISA. TC:PC ratio refers to relative amounts of TC and PC plasmids included in the transfection. Numbers above the bars represent fold induction (See FIG. 1). See FIG. 3 for measurement details.

(13) FIG. 13. Investigating the IL-2 producing subset of Jurkat T cells. Jurkat T cells were transfected with VEGF-MESA at a TC:PC ratio of 10 and sgRNA 7 and treated with VEGF and Brefeldin A (the latter blocks protein secretion), for 24 h. IL-2 production was evaluated by labeling intracellular IL-2 in fixed cells utilizing an anti-IL-2 antibody and then co-evaluating both IL-2 labeling and expression of a transfection control (BFP) by flow cytometry. Each sample was analyzed by dividing cells into two subsets based upon the expression (or non-expression) of the transfection control (BFP), and the mean level of anti-IL-2 labeling (APC) was quantified. Each sample was conducted in biological triplicate, and error bars represent one standard deviation. (* $p \leq 0.001$).

DETAILED DESCRIPTION

(14) The present invention is described herein using several definitions, as set forth below and throughout the application.

(15) Unless otherwise specified or indicated by context, the terms “a”, “an”, and “the” mean “one or more.” For example, “a receptor,” “ligand,” and “complex” should be interpreted to mean “one or more receptors,” “one or more ligands,” and “one or more complexes,” respectively.

(16) As used herein, “about,” “approximately,” “substantially,” and “significantly” will be understood by persons of ordinary skill in the art and will vary to some extent on the context in which they are used. If there are uses of these terms which are not clear to persons of ordinary skill in the art given the context in which they are used, “about” and “approximately” will mean plus or minus $\leq 10\%$ of the particular term and “substantially” and “significantly” will mean plus or

minus>10% of the particular term.

(17) As used herein, the terms “include” and “including” have the same meaning as the terms “comprise” and “comprising” in that these latter terms are “open” transitional terms that do not limit claims only to the recited elements succeeding these transitional terms. The term “consisting of,” while encompassed by the term “comprising,” should be interpreted as a “closed” transitional term that limits claims only to the recited elements succeeding this transitional term. The term “consisting essentially of,” while encompassed by the term “comprising,” should be interpreted as a “partially closed” transitional term which permits additional elements succeeding this transitional term, but only if those additional elements do not materially affect the basic and novel characteristics of the claim.

(18) The disclosed technology relates to “extracellular sensors.” As disclosed herein, an “extracellular sensor” is a molecule or a system of molecules that can be used to bind to a ligand and provide a detectable response based on binding the ligand. In some cases, “extracellular sensor” may be referred to as “molecular switches.” Extracellular sensors are disclosed in the art. (See, e.g., Daringer et al., “Modular Extracellular Sensor Architecture for Engineering Mammalian Cell-based Devices,” Nichole M. Daringer, Rachel M. Dudek, Kelly A. Schwarz, and Josh N. Leonard, ACS Synth. Biol. 2014, 3, 892-902, published Feb. 25, 2014; WO 2013/022739, published on Feb. 14, 2013; and U.S. Publication No. 2014-0234851; the contents of which are incorporated herein by reference in their entireties).

(19) As contemplated herein, a “ligand-binding protein” is a macromolecule, typically a protein, which binds to a ligand. For example, a ligand-binding protein may include a receptor for a ligand or a portion of a receptor for a ligand, for example, where the receptor is a membrane protein and the ligand-binding protein comprises the extracellular portion of the receptor that binds an extracellular ligand.

(20) The disclosed extracellular sensors may be utilized for sensing an extracellular ligand and providing a molecular signal to a cell when the ligand is sensed. The molecular signal may result in modulating expression from an endogenous promoter in the cell or an exogenous promoter in the cell. As used herein, “modulating expression” may include “activating expression” or “de-repressing expression,” and/or “modulating expression” may include “repressing expression” and/or “inhibiting expression.”

(21) Reference is made herein to nucleic acid and nucleic acid sequences. The terms “nucleic acid” and “nucleic acid sequence” refer to a nucleotide, oligonucleotide, polynucleotide (which terms may be used interchangeably), or any fragment thereof. These phrases also refer to DNA or RNA of genomic or synthetic origin (which may be single-stranded or double-stranded and may represent the sense or the antisense strand).

(22) Reference also is made herein to peptides, polypeptides, proteins and compositions comprising peptides, polypeptides, and proteins. As used herein, a polypeptide and/or protein is defined as a polymer of amino acids, typically of length ≥ 100 amino acids (Garrett & Grisham, Biochemistry, 2nd edition, 1999, Brooks/Cole, 110). A peptide is defined as a short polymer of amino acids, of a length typically of 20 or less amino acids, and more typically of a length of 12 or less amino acids (Garrett & Grisham, Biochemistry, 2nd edition, 1999, Brooks/Cole, 110).

(23) As disclosed herein, exemplary peptides, polypeptides, proteins may comprise, consist essentially of, or consist of any reference amino acid sequence disclosed herein, or variants of the peptides, polypeptides, and proteins may comprise, consist essentially of, or consist of an amino acid sequence having at least about 80%, 90%, 95%, 96%, 97%, 98%, or 99% sequence identity to any amino acid sequence disclosed herein, for example SEQ ID NOs:1-17. Variant peptides, polypeptides, and proteins may include peptides, polypeptides, and proteins having one or more amino acid substitutions, deletions, additions and/or amino acid insertions relative to a reference peptide, polypeptide, or protein. Also disclosed are nucleic acid molecules that encode the disclosed peptides, polypeptides, and proteins (e.g., polynucleotides that encode any of the

peptides, polypeptides, and proteins disclosed herein and variants thereof).

(24) The term “amino acid,” includes but is not limited to amino acids contained in the group consisting of alanine (Ala or A), cysteine (Cys or C), aspartic acid (Asp or D), glutamic acid (Glu or E), phenylalanine (Phe or F), glycine (Gly or G), histidine (His or H), isoleucine (Ile or I), lysine (Lys or K), leucine (Leu or L), methionine (Met or M), asparagine (Asn or N), proline (Pro or P), glutamine (Gln or Q), arginine (Arg or R), serine (Ser or S), threonine (Thr or T), valine (Val or V), tryptophan (Trp or W), and tyrosine (Tyr or Y) residues. The term “amino acid residue” also may include amino acid residues contained in the group consisting of homocysteine, 2-Aminoadipic acid, N-Ethylasparagine, 3-Aminoadipic acid, Hydroxylysine, β -alanine, β -Amino-propionic acid, allo-Hydroxylysine acid, 2-Aminobutyric acid, 3-Hydroxyproline, 4-Aminobutyric acid, 4-Hydroxyproline, piperidinic acid, 6-Aminocaproic acid, Isodesmosine, 2-Aminoheptanoic acid, allo-Isoleucine, 2-Aminoisobutyric acid, N-Methylglycine, sarcosine, 3-Aminoisobutyric acid, N-Methylisoleucine, 2-Aminopimelic acid, 6-N-Methyllysine, 2,4-Diaminobutyric acid, N-Methylvaline, Desmosine, Norvaline, 2,2'-Diaminopimelic acid, Norleucine, 2,3-Diaminopropionic acid, Ornithine, and N-Ethylglycine. Typically, the amide linkages of the peptides are formed from an amino group of the backbone of one amino acid and a carboxyl group of the backbone of another amino acid.

(25) The amino acid sequences contemplated herein may include conservative amino acid substitutions relative to a reference amino acid sequence. For example, a variant peptides, polypeptides, and proteins as contemplated herein may include conservative amino acid substitutions relative to an amino acid sequence of a reference peptide, polypeptide, or protein. “Conservative amino acid substitutions” are those substitutions that are predicted to interfere least with the properties of the reference peptide, polypeptide, or protein. In other words, conservative amino acid substitutions substantially conserve the structure and the function of the reference peptide, polypeptide, or protein. The following table provides a list of exemplary conservative amino acid substitutions.

(26) TABLE-US-00001 Table of Conservative Amino Acid Substitutions

Original	Conservative Residue Substitution
Ala	Gly, Ser
Arg	His, Lys
Asn	Asp, Gln
Asp	Glu, His
Glu	Asp, Glu
Gln	Asn, Glu
His	Asp, Glu
Ile	Val, Leu
Leu	Val, Ile
Lys	Arg, His
Met	Leu, Ile
Phe	Tyr, Trp
Trp	Tyr, Phe
Tyr	Phe, Trp
Val	Ile, Leu

(27) “Non-conservative amino acid substitutions” are those substitutions that are predicted to interfere most with the properties of the reference peptide, polypeptide, or protein. For example, a non-conservative amino acid substitution might replace a basic amino acid at physiological pH such as Arg, His, or Lys, with a non-basic or acidic amino acid at physiological pH such as Asp or Glu. A non-conservative amino acid substitution might replace a non-polar amino acid at physiological pH such as Ala, Gly, Ile, Leu, Phe, or Val, with a polar amino acid at physiological pH such as Arg, Asp, Glu, His, or Lys.

(28) The peptides, polypeptides, and proteins disclosed herein may be modified to include non-amino acid moieties. Modifications may include but are not limited to carboxylation (e.g., N-terminal carboxylation via addition of a di-carboxylic acid having 4-7 straight-chain or branched carbon atoms, such as glutaric acid, succinic acid, adipic acid, and 4,4-dimethylglutaric acid), amidation (e.g., C-terminal amidation via addition of an amide or substituted amide such as alkylamide or dialkylamide), PEGylation (e.g., N-terminal or C-terminal PEGylation via additional of polyethylene glycol), acylation (e.g., O-acylation (esters), N-acylation (amides), S-acylation (thioesters)), acetylation (e.g., the addition of an acetyl group, either at the N-terminus of the protein or at lysine residues), formylation lipoylation (e.g., attachment of a lipoate, a C8 functional group), myristoylation (e.g., attachment of myristate, a C14 saturated acid), palmitoylation (e.g., attachment of palmitate, a C16 saturated acid), alkylation (e.g., the addition of an alkyl group, such as an methyl at a lysine or arginine residue), isoprenylation or prenylation (e.g., the addition of an

isoprenoid group such as farnesol or geranylgeraniol), amidation at C-terminus, glycosylation (e.g., the addition of a glycosyl group to either asparagine, hydroxylysine, serine, or threonine, resulting in a glycoprotein). Distinct from glycation, which is regarded as a nonenzymatic attachment of sugars, polysialylation (e.g., the addition of polysialic acid), glypiation (e.g., glycosylphosphatidylinositol (GPI) anchor formation, hydroxylation, iodination (e.g., of thyroid hormones), and phosphorylation (e.g., the addition of a phosphate group, usually to serine, tyrosine, threonine or histidine).

(29) Variants comprising deletions relative to a reference amino acid sequence or nucleotide sequence are contemplated herein. A “deletion” refers to a change in the amino acid or nucleotide sequence that results in the absence of one or more amino acid residues or nucleotides relative to a reference sequence. A deletion removes at least 1, 2, 3, 4, 5, 10, 20, 50, 100, or 200 amino acids residues or nucleotides. A deletion may include an internal deletion or a terminal deletion (e.g., an N-terminal truncation or a C-terminal truncation or both of a reference polypeptide or a 5'-terminal or 3'-terminal truncation or both of a reference polynucleotide).

(30) Variants comprising a fragment of a reference amino acid sequence or nucleotide sequence are contemplated herein, for example fragments of an of SEQ ID NOs:1-17. A “fragment” is a portion of an amino acid sequence or a nucleotide sequence which is identical in sequence to but shorter in length than the reference sequence. A fragment may comprise up to the entire length of the reference sequence, minus at least one nucleotide/amino acid residue. For example, a fragment may comprise from 5 to 1000 contiguous nucleotides or contiguous amino acid residues of a reference polynucleotide or reference polypeptide, respectively. In some embodiments, a fragment may comprise at least 5, 10, 15, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 40, 50, 60, 70, 80, 90, 100, 150, 250, or 500 contiguous nucleotides or contiguous amino acid residues of a reference polynucleotide or reference polypeptide, respectively. Fragments may be preferentially selected from certain regions of a molecule, for example the N-terminal region and/or the C-terminal region of a polypeptide or the 5'-terminal region and/or the 3' terminal region of a polynucleotide. The term “at least a fragment” encompasses the full length polynucleotide or full length polypeptide.

(31) Variants comprising insertions or additions relative to a reference sequence are contemplated herein. The words “insertion” and “addition” refer to changes in an amino acid or nucleotide sequence resulting in the addition of one or more amino acid residues or nucleotides. An insertion or addition may refer to 1, 2, 3, 4, 5, 10, 20, 30, 40, 50, 60, 70, 80, 90, 100, 150, or 200 amino acid residues or nucleotides.

(32) Fusion proteins and fusion polynucleotides also are contemplated herein. The disclosed extracellular sensors may be defined as a fusion protein. A “fusion protein” refers to a protein formed by the fusion of at least one peptide, polypeptide, protein or variant thereof as disclosed herein to at least one molecule of a heterologous peptide, polypeptide, protein or variant thereof. The heterologous protein(s) may be fused at the N-terminus, the C-terminus, or both termini. A fusion protein comprises at least a fragment or variant of the heterologous protein(s) that are fused with one another, preferably by genetic fusion (i.e., the fusion protein is generated by translation of a nucleic acid in which a polynucleotide encoding all or a portion of a first heterologous protein is joined in-frame with a polynucleotide encoding all or a portion of a second heterologous protein). The heterologous protein(s), once part of the fusion protein, may each be referred to herein as a “portion”, “region” or “moiety” of the fusion protein. For example, where the fusion protein comprises at least a portion of a ligand binding protein, at least a portion of a transmembrane protein, at least a portion of a protease, the portions of the fusion protein may be referred to as “a ligand binding portion,” “a transmembrane portion,” and “a protease portion,” respectively.

(33) A fusion polynucleotide refers to the fusion of the nucleotide sequence of a first polynucleotide to the nucleotide sequence of a second heterologous polynucleotide (e.g., the 3' end of a first polynucleotide to a 5' end of the second polynucleotide). Where the first and second polynucleotides encode proteins, the fusion may be such that the encoded proteins are in-frame and

results in a fusion protein. The first and second polynucleotide may be fused such that the first and second polynucleotide are operably linked (e.g., as a promoter and a gene expressed by the promoter as discussed below).

(34) “Homology” refers to sequence similarity or, interchangeably, sequence identity, between two or more polypeptide sequences or polynucleotide sequences. Homology, sequence similarity, and percentage sequence identity may be determined using methods in the art and described herein.

(35) The phrases “percent identity” and “% identity,” as applied to polypeptide sequences, refer to the percentage of residue matches between at least two polypeptide sequences aligned using a standardized algorithm. Methods of polypeptide sequence alignment are well-known. Some alignment methods take into account conservative amino acid substitutions. Such conservative substitutions, explained in more detail above, generally preserve the charge and hydrophobicity at the site of substitution, thus preserving the structure (and therefore function) of the polypeptide. Percent identity for amino acid sequences may be determined as understood in the art. (See, e.g., U.S. Pat. No. 7,396,664, which is incorporated herein by reference in its entirety). A suite of commonly used and freely available sequence comparison algorithms is provided by the National Center for Biotechnology Information (NCBI) Basic Local Alignment Search Tool (BLAST) (Altschul, S. F. et al. (1990) *J. Mol. Biol.* 215:403-410), which is available from several sources, including the NCBI, Bethesda, Md., at its website. The BLAST software suite includes various sequence analysis programs including “blastp,” that is used to align a known amino acid sequence with other amino acids sequences from a variety of databases.

(36) Percent identity may be measured over the length of an entire defined polypeptide sequence or may be measured over a shorter length, for example, over the length of a fragment taken from a larger, defined polypeptide sequence, for instance, a fragment of at least 15, at least 20, at least 30, at least 40, at least 50, at least 70 or at least 150 contiguous residues. Such lengths are exemplary only, and it is understood that any fragment length may be used to describe a length over which percentage identity may be measured.

(37) A “variant” of a particular polypeptide sequence may be defined as a polypeptide sequence having at least 50% sequence identity to the particular polypeptide sequence over a certain length of one of the polypeptide sequences using blastp with the “BLAST 2 Sequences” tool available at the National Center for Biotechnology Information's website. (See Tatiana A. Tatusova, Thomas L. Madden (1999), “Blast 2 sequences—a new tool for comparing protein and nucleotide sequences”, *FEMS Microbiol Lett.* 174:247-250). In some embodiments a variant polypeptide may show, for example, at least 60%, at least 70%, at least 80%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% or greater sequence identity over a certain defined length relative to a reference polypeptide such as a polypeptide comprising an amino acid sequence of any of SEQ ID NOs:1-17.

(38) A variant polypeptide may have substantially the same functional activity as a reference polypeptide. For example, a variant polypeptide may exhibit or more biological activities associated with binding a ligand and/or binding DNA at a specific binding site.

(39) The terms “percent identity” and “% identity,” as applied to polynucleotide sequences, refer to the percentage of residue matches between at least two polynucleotide sequences aligned using a standardized algorithm. Such an algorithm may insert, in a standardized and reproducible way, gaps in the sequences being compared in order to optimize alignment between two sequences, and therefore achieve a more meaningful comparison of the two sequences. Percent identity for a nucleic acid sequence may be determined as understood in the art. (See, e.g., U.S. Pat. No. 7,396,664, which is incorporated herein by reference in its entirety). A suite of commonly used and freely available sequence comparison algorithms is provided by the National Center for Biotechnology Information (NCBI) Basic Local Alignment Search Tool (BLAST) (Altschul, S. F. et al. (1990) *J. Mol. Biol.* 215:403-410), which is available from several sources, including the NCBI, Bethesda, Md., at its website. The BLAST software suite includes various sequence analysis

programs including “blastn,” that is used to align a known polynucleotide sequence with other polynucleotide sequences from a variety of databases. Also available is a tool called “BLAST 2 Sequences” that is used for direct pairwise comparison of two nucleotide sequences. “BLAST 2 Sequences” can be accessed and used interactively at the NCBI website. The “BLAST 2 Sequences” tool can be used for both blastn and blastp (discussed above).

(40) Percent identity may be measured over the length of an entire defined polynucleotide sequence or may be measured over a shorter length, for example, over the length of a fragment taken from a larger, defined sequence, for instance, a fragment of at least 20, at least 30, at least 40, at least 50, at least 70, at least 100, or at least 200 contiguous nucleotides. Such lengths are exemplary only, and it is understood that any fragment length may be used to describe a length over which percentage identity may be measured.

(41) A “full length” polynucleotide sequence is one containing at least a translation initiation codon (e.g., methionine) followed by an open reading frame and a translation termination codon. A “full length” polynucleotide sequence encodes a “full length” polypeptide sequence.

(42) A “variant,” “mutant,” or “derivative” of a particular nucleic acid sequence may be defined as a nucleic acid sequence having at least 50% sequence identity to the particular nucleic acid sequence over a certain length of one of the nucleic acid sequences using blastn with the “BLAST 2 Sequences” tool available at the National Center for Biotechnology Information's website. (See Tatiana A. Tatusova, Thomas L. Madden (1999), “Blast 2 sequences—a new tool for comparing protein and nucleotide sequences”, *FEMS Microbiol Lett.* 174:247-250). In some embodiments a variant polynucleotide may show, for example, at least 60%, at least 70%, at least 80%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% or greater sequence identity over a certain defined length relative to a reference polynucleotide.

(43) Nucleic acid sequences that do not show a high degree of identity may nevertheless encode similar amino acid sequences due to the degeneracy of the genetic code. It is understood that changes in a nucleic acid sequence can be made using this degeneracy to produce multiple nucleic acid sequences that all encode substantially the same protein.

(44) “Operably linked” refers to the situation in which a first nucleic acid sequence is placed in a functional relationship with a second nucleic acid sequence. For instance, a promoter is operably linked to a coding sequence if the promoter affects the transcription or expression of the coding sequence. Operably linked DNA sequences may be in close proximity or contiguous and, where necessary to join two protein coding regions, in the same reading frame.

(45) A “recombinant nucleic acid” is a sequence that is not naturally occurring or has a sequence that is made by an artificial combination of two or more otherwise separated segments of sequence. This artificial combination is often accomplished by chemical synthesis or, more commonly, by the artificial manipulation of isolated segments of nucleic acids, e.g., by genetic engineering techniques such as those described in Sambrook, J. et al. (1989) *Molecular Cloning: A Laboratory Manual*, 2nd ed., vol. 1 3, Cold Spring Harbor Press, Plainview N.Y. The term recombinant includes nucleic acids that have been altered solely by addition, substitution, or deletion of a portion of the nucleic acid. Frequently, a recombinant nucleic acid may include a nucleic acid sequence operably linked to a promoter sequence. Such a recombinant nucleic acid may be part of a vector that is used, for example, to transform a cell.

(46) “Transformation” describes a process by which exogenous DNA is introduced into a recipient cell. Transformation may occur under natural or artificial conditions according to various methods well known in the art, and may rely on any known method for the insertion of foreign nucleic acid sequences into a prokaryotic or eukaryotic host cell. The method for transformation is selected based on the type of host cell being transformed and may include, but is not limited to, bacteriophage or viral infection, electroporation, heat shock, lipofection, and particle bombardment. The term “transformed cells” includes stably transformed cells in which the inserted

DNA is capable of replication either as an autonomously replicating plasmid or as part of the host chromosome, as well as transiently transformed cells which express the inserted DNA or RNA for limited periods of time.

(47) A “composition comprising a given polypeptide” and a “composition comprising a given polynucleotide” refer broadly to any composition containing the given polynucleotide or amino acid sequence. The composition may comprise a dry formulation or an aqueous solution. The compositions may be stored in any suitable form including, but not limited to, freeze-dried form and may be associated with a stabilizing agent such as a carbohydrate. The compositions may be aqueous solution containing salts (e.g., NaCl), detergents (e.g., sodium dodecyl sulfate; SDS), and other components.

(48) “Substantially isolated or purified” nucleic acid or amino acid sequences are contemplated herein. The term “substantially isolated or purified” refers to nucleic acid or amino acid sequences that are removed from their natural environment, and are at least 60% free, preferably at least 75% free, and more preferably at least 90% free, even more preferably at least 95% free from other components with which they are naturally associated.

(49) Antibodies, antigen-binding fragments of antibodies, and fusion proteins comprising fragments of antibodies are contemplated herein. For example, the ligand binding domain of the disclosed exogenous extracellular sensors may comprise an antibody, an antigen-binding fragment of an antibody, or a fusion protein comprising fragments of an antibody, where the antibody, the antigen-binding fragment of the antibody, or the fusion protein comprising fragments of an antibody binds to a ligand. Suitable antibodies, antigen-binding fragments thereof, and fusion proteins comprising fragments of an antibody may include, but are not limited to Fab fragments, single-chain variable fragments (scFvs), and single-domain antibodies (sdAb or nanobodies).

(50) The disclosed sensors include a protease cleavage sequence. The sensors are not limited to any particular protease or corresponding protease cleavage site. In some embodiments, the protease and cleavage site are from a virus. For example, in certain embodiments, the protease and protease cleavage site are from a virus selected from: tobacco etch virus (TEV), a chymotrypsin-like serine protease and corresponding cleavage sites, alphavirus proteases and cleavage sites, Hepatitis C virus proteases (e.g., NS3 proteases) and corresponding cleavage sites, chymotrypsin-like cysteine proteases and corresponding cleavage sites, papain-like cysteine proteases and cleavage sites, picornavirus leader proteases and cleavage sites, HIV proteases and cleavage sites, Herpesvirus proteases and cleavage sites, and adenovirus proteases and cleavage sites (see, Tong, Chem. Rev. 2002, 102, 4609-4626, herein incorporated by reference in its entirety). In particular embodiments, the proteases and cleavage sites are bacterial in origin, such as, for example, from *Streptomyces griseus* protease A (SGPA), SGPB, and alpha-lytic protease and corresponding cleavage sites. In some embodiments, the proteases and cleavage sites are mammalian. For example, the proteases could be one of the five major classes of proteases known in mammals which include serine proteases, cysteine proteases, metallo proteases, aspartic proteases, and threonine proteases (see, e.g., Turk et al., The EMBO Journal, 2012, 31, 1630-1643; Lopez-Otin and Overall, 2002, Nat. Rev. Mol. Cell Biol., 2:509-519; Overall and Blobel, 2007, Nat. Rev. Mol. Cell Biol., 8: 245-257; and Lopez-Otin and Bond, 2008, J. Biol. Chem., 283:30422-30437, all of which are herein incorporated in their entireties by references).

(51) The disclosed exogenous extracellular sensors may be utilized to modulate expression of endogenous or exogenous genes in the presence of an extracellular ligand for the sensors. For example, the functional domain of the exogenous extracellular sensors may be targeted to a promoter region of an endogenous or exogenous gene to modulate expression of the gene. In some embodiments, the functional domain contains a transcription modulator that induces expression from a targeted promoter. In other embodiments, the functional domain contains a transcription modulator that inhibits expression from a targeted promoter. In further embodiments, the functional domain contains a subdomain that modulates the chromatin state of a targeted promoter.

(52) The functional domain of the exogenous extracellular sensors may be targeted to an endogenous or exogenous gene via an RNA-binding subdomain and a targeting RNA. Suitable RNA-binding subdomains may include, but are not limited to, RNA-binding subdomains of bacterial proteins, such as Cas9 proteins, and suitable RNAs may include, but are not limited to, sgRNAs. Suitable RNA-binding subdomain of Cas9 proteins may include RNA-binding subdomains of Cas9 protein of *Streptococcus pyogenes* (SEQ ID NO:1) *Neisseria meningitidis* (SEQ ID NO:2), *Streptococcus thermophilus* (SEQ ID NO:3), *Treponema denticola* (SEQ ID NO:4), *Staphylococcus aureus* (SEQ ID NO: 5), *Brevibacillus laterosporus* (SEQ ID NO:6), *Campylobacter jejuni* (SEQ ID NO:7), *Listeria monocytogenes* (SEQ ID NO:8), *Lactobacillus salivarius* (SEQ ID NO:9), *Legionella pneumophila* (SEQ ID NO:10), *Francisella novicida* (SEQ ID NO:11) and the like. A suitable sgRNA may be selected and/or designed based on the RNA-binding subdomain of the functional domain and a selected target gene. For a sgRNA may be designed to bind to the RNA-binding subdomain of the functional domain and target the functional domain to a target gene, which may be an endogenous or exogenous target gene.

ILLUSTRATIVE EMBODIMENTS

(53) The following embodiments are illustrative and do not limit the scope of the claimed subject matter.

(54) Embodiment 1. An exogenous extracellular sensor and/or a nucleic acid sequence encoding the exogenous extracellular sensor, wherein the exogenous extracellular sensor comprises: a) a ligand binding domain; b) a transmembrane domain; c) a protease cleavage site; and d) a functional domain comprising an RNA-binding subdomain fused to a transcription regulatory subdomain (e.g., a regulatory subdomain that activates transcription from a promoter, a regulatory subdomain that repressed transcription from a promoter, and/or a regulatory subdomain that modulates chromatin state).

(55) Embodiment 2. The exogenous extracellular sensor of embodiment 1, wherein the RNA-binding subdomain comprises a RNA-binding domain of a Cas9 protein.

(56) Embodiment 3. The exogenous extracellular sensor of embodiment 1 or 2, wherein the exogenous extracellular sensor further comprises an extracellular spacer.

(57) Embodiment 4. The exogenous extracellular sensor of any of the foregoing embodiments, wherein the exogenous extracellular sensor further comprises an intracellular spacer that is one, two, three, four, five, or six amino acids in length.

(58) Embodiment 5. The exogenous extracellular sensor of any of the foregoing embodiments, wherein the transmembrane domain of is immediately adjacent to the protease cleavage site such that there is no intracellular spacer there between.

(59) Embodiment 6. A system comprising the exogenous extracellular sensor of any of embodiments 1-5 as a first exogenous extracellular sensor and further comprising a second exogenous extracellular sensor, wherein the second exogenous extracellular sensor comprises: e) a ligand binding domain, f) a transmembrane domain, and g) a protease domain that cleaves the protease cleavage site of the first exogenous extracellular receptor.

(60) Embodiment 7. The system of embodiment 6 further comprising a cell wherein the first and second exogenous extracellular sensors are present in the cell as cell membrane proteins.

(61) Embodiment 8. The system of embodiment 7, wherein the first and second exogenous sensors are located in the cell membrane such that the first and second ligand binding domains are located outside the cell and the protease cleavage site and the functional domain are located inside the cell.

(62) Embodiment 9. The system of embodiment 8, wherein the first and second ligand binding domains bind the same ligand, and wherein the first and second exogenous sensors are configured such that the protease domain will cleave the protease cleavage site when the first and second ligand binding domains bind the same ligand.

(63) Embodiment 10. The system of any of embodiments 7-9 further comprising a RNA or a vector that expresses the RNA wherein the RNA-binding subdomain of the functional domain binds the

RNA.

(64) Embodiment 11. The system of embodiment 10, wherein the RNA is a sgRNA or a vector that expresses the sgRNA wherein the RNA-binding subdomain of the functional domain binds the sgRNA and the functional domain is targeted to an endogenous gene of the cell or an exogenous gene introduced to the cell.

(65) Embodiment 12. The system of embodiment 10 or 11, wherein the RNA-binding domain of the functional domain comprises a Cas9 protein domain.

(66) Embodiment 13. The system of any of embodiment 6-12 further comprising a genetic construct, wherein the genetic construct is configured to express a gene in response to the transcription regulatory domain.

(67) Embodiment 14. The system of any of embodiments 6-13, wherein the protease is at least partially auto-inhibited.

(68) Embodiment 15. A method for modulating expression of an endogenous gene of a cell, the method comprising introducing the system of embodiment 11 to the cell and contacting the first and second ligand binding domains with the ligand for the first and second ligand binding domains wherein the functional domain is targeted to the endogenous gene of the cell and modulates expression of the endogenous gene of the cell in the presence of the ligand for the first and second ligand binding domains.

EXAMPLES

(69) The following Examples are illustrative and are not intended to limit the scope of the claimed subject matter.

Example 1

(70) Reference is made to the invention disclosure entitled “Modular Extracellular Sensor Architecture for Regulating Endogenous Genes,” Josh N. Leonard, and Kelly A. Schwarz, submitted Jul. 22, 2015.

(71) Title: Modular Extracellular Sensor Architecture for Regulating Endogenous Genes

TECHNICAL FIELD

(72) The technology of this example comprises a general platform for building mammalian cell-based therapies and devices that interface robustly with host physiology. Specifically, this technology leverages our Modular Extracellular Sensor Architecture (MESA), a system for engineering novel receptor proteins that confer the detection of extracellular cues. (See Example 5). When MESA receptors bind an extracellular analyte, this “sensing” event is translated into the intracellular release of a previously sequestered protein, such as a transcription factor. In this example, we disclose a novel class of MESA receptor in which the sensing event releases a readily programmable protein based upon the CRISPR Cas9 protein. Cas9 is a microbial nuclease that can be programmed to bind to a DNA locus of interest via co-expression with a small guide RNA (sgRNA). Variants of Cas9 have previously been developed such that programmed DNA binding confers gene editing, transcriptional silencing, or transcriptional activation, and other modalities may be possible. This invention thus enables one to engineer a cell such that the detection of essentially any extracellular species is translated into the regulation (in various ways) of any one or more endogenous genes. This novel capability vastly expands our ability to engineer cells as diagnostics, therapeutics, and tools for fundamental biology research.

(73) The applications of the technology of this example may include but are not limited to: engineered cell-based therapeutics: cells that sense molecules in their environment and respond via release of a therapeutic payload (e.g., custom-engineering immune cells that release immune potentiating factors only in the vicinity of tumors, serving as targeted in situ factories); cell-based diagnostics tools and assays enabling the detection and monitoring of specific molecules or analytes; and customized research reagents that investigators may use to functionally “rewire” their system of interest, comprising a type of experimental perturbation not possible via existing drugs and gene knockout approaches.

(74) The advantages of the technology of this example may include but are not limited to: expansion of the capabilities of MESA by enabling novel capabilities including: (1) readily programmable choice of MESA “output” genes via simple DNA sequence analysis (rather than protein engineering), (2) regulation of endogenous genes, and (3) combinatorial regulation of endogenous genes; and Cas9-based gene regulation conditional upon the detection of a specific, extracellular cue.

BRIEF SUMMARY

(75) The technology of this example comprises a general platform for building mammalian cell-based therapies and devices that interface robustly with host physiology. Specifically, this technology leverages our Modular Extracellular Sensor Architecture (MESA), a system for engineering novel receptor proteins that confer the detection of extracellular cues. When MESA receptors bind an extracellular analyte, this “sensing” event is translated into the intracellular release of a previously sequestered protein, such as a transcription factor. MESA receptors comprise of a pair of synthetic protein receptor chains that dimerize in the presence of a ligand, enabling the protease on the protease chain (PC) to cut a cognate cleavage sequence (CS) on the target chain (TC), thus releasing a tethered transcription factor. Previously described MESA receptors release engineered transcription factors such as the tet transactivator (tTA), which binds one specific DNA sequence. Thus, MESA receptors have been used thus far only to regulate engineered promoters, and engineering novel transcriptional “outputs” required engineering novel transcription factor proteins with DNA binding specificity customized to the novel target sequence. This invention comprises engineering the TC of a MESA receptor to include a tethered version of a CRISPR Cas9-based protein. Natural Cas9 is a nuclease that hones to specific DNA sequences via interaction with a small guide RNA (sgRNA) that is complementary to the target DNA sequence of interest. Variants of Cas9 have been developed such that programmed DNA binding confers gene editing, transcriptional silencing, or transcriptional activation, and other modalities may be possible. Because sgRNAs are easy to design using extant bioinformatic tools, this invention enables one to engineer novel MESA receptors (those that regulate one or more endogenous genes) in an expedient fashion that circumvents the need for slow, costly, difficult, protein engineering.

Technical Description

(76) We have developed and validated a MESA receptor that releases a Cas9-based transcription activator from the target chain upon ligand binding. (See FIG. 1). This catalytically inactive Cas9 (dCas9) was fused to VP64 (dCas9-VP64) to create a transcriptional activator, although in principle any such Cas9-based protein could be similarly incorporated into MESA. For example, fusion of dCas9 to a transcriptional repressive protein, such as KRAB, has been reported to generate transcriptional repressors; a MESA receptor including such a protein would be expected to confer gene repression in a manner that is conditional upon sensing of extracellular cues. Our initial demonstrations have utilized the dCas9-VP64 approach to confer conditional transcriptional activation of target genes.

(77) In a proof-of-concept experiment, we generated sgRNAs targeting the promoter of a reporter construct (a DNA sequence in which transcriptional activation leads to expression of a fluorescent protein), and we showed that a MESA-dCas9-VP64 receptor conferred ligand-inducible expression of the reporter gene. (See FIG. 2.) We then demonstrated that it was also possible to activate endogenous genes via MESA. By expressing MESA-dCas9-VP64 along with sgRNAs targeting the endogenous human IL-2 promoter, we observed that upon addition of ligand, significant increases in IL-2 gene expression were induced. (See FIG. 3). Furthermore, we demonstrated that novel Cas9 technologies can be readily incorporated into Cas9-based MESA; it was recently reported that dCas9-based transcriptional activators may be potentiated by co-expression of an engineered protein that recruits multiple transcriptional activation domains (P65 and HSF1) to the dCas9-VP64-sgRNA complex. We observed that this strategy also enhanced MESA-dCas9-VP64-mediated transcriptional activation, emphasizing the power of this invention's modular.

(78) The technology of this example enables one to rapidly engineer novel receptor proteins that regulate one or more endogenous genes in a readily defined fashion. Such a technology meets an unmet need in the general field of cell-based therapies. For example, engineering local, sustained production of immune modulating factors is a widely-acknowledged need in the field of cancer immunotherapy; such a capability could complement and vastly expand the range of cancers amenable to treatment using the rapidly-expanding commercial field of engineered cell-based cancer therapies. Ultimately this invention could improve both the safety and efficacy of such immunotherapy approaches. An additional commercial opportunity comprises the provision of customized reagents for fundamental biological research. Much like genetic knockout mice have proven to be an invaluable tool for studying organismal biology and using such model animals for translational research, this invention would enable researchers to “rewire” organismal biology in a manner not currently possible. Such experimental perturbations are therefore uniquely valuable, and much like experimental animal lines, could be commercialized as customized products for basic research (in various potential commercial models, such as design and provision of customized reagents and services).

(79) The technology of this example could enable a range of new therapeutic capabilities including therapeutic applications such as cancer immunotherapy. The technology of this example also relates to a novel class of experimental reagent with wide-ranging potential utility.

Example 2

(80) Reference is made to the Abstract entitled “Rewiring Cellular Input-Output to Engineer Cell-Based Therapies That Interface Robustly with Human Physiology,” Rachel M. Dudek, Nichole M. Daringer, and Joshua N. Leonard,” presented at the 2015 AIChE Annual Meeting, Session: Emerging Frontiers in Systems and Synthetic Biology, Nov. 8-13, 2015, which is incorporated by reference herein in its entirety.

(81) Title: Rewiring Cellular Input-Output to Engineer Cell-Based Therapies that Interface Robustly with Human Physiology

(82) Cell-based therapies have proven to be useful for treating a wide variety of diseases including autoimmune disease, infectious diseases, and cancer. A particularly promising frontier is the use of engineered cell therapies, wherein cells are programmed to carry out custom functions such as harnessing the immune system to find and destroy cancerous cells. Such therapies have now achieved robust clinical successes for some applications. However, controlling or modulating the activity of these therapies post-implantation remains both attractive, to enhance both safety and therapeutic efficacy, and challenging, since the tools necessary to engineer custom functional programs are limited. In particular, engineering cells to sense and recognize specific combinations of environmental cues would be desirable, but no such technology has yet been reported. Toward this goal, our lab has developed a platform for engineering novel protein biosensors, termed modular extracellular sensor architecture (MESA), for detecting exclusively extracellular cues. Upon binding extracellular ligand, MESA receptors release sequestered transcription factors from the cytoplasmic face of the plasma membrane, freeing the factor to regulate expression of an “output” gene or genes within the cell. Here we present three areas in which we have recently expanded upon the capabilities of the core MESA technology: (1) achieving novel ligand recognition by integrating modular ligand-binding domains, (2) integrating MESA with intracellular gene circuits that enable the cell to “process” sensory information by logical evaluation, and (3) rewiring cellular input-output by coupling extracellular sensing to intracellular modulation of endogenous gene expression.

(83) First, as modular ligand binding domains, we used camelid single domain antibody fragments termed “nanobodies”. Each nanobody's small size enables multiple nanobodies to recognize distinct, non-overlapping epitopes on a ligand of interest. Thus nano body MESA are able to recognize monomeric or asymmetric ligand inputs. After demonstrating the feasibility of nano body MESA, we integrated multiple such receptors into a logical processing circuit that enables the

multiplexed logical evaluation of distinct protein ligand “inputs”. Next, we demonstrated that human antibody-derived scFv could also be used as ligand-binding domains in MESA, and VEGF-MESA were thus designed to sense the tumor-associated cytokine, VEGF. In order to regulate endogenous genes, the MESA transcription factor domain was replaced with the dCas9 protein, which is able to act as a ‘pseudo-transcription factor’ by tethering transactivation domains to the dCas9 protein and providing small guide RNAs in trans to direct the Cas9 protein to a specific genomic locus. As proof of principle, we developed VEGF-MESA that induce expression of IL-2, a cytokine important for T-cell growth and proliferation. In principle, the dCas9 protein can be guided to any genomic locus of interest by simply delivering a different small guide RNA. Cells expressing VEGF-MESA exhibited minimal background signaling in the absence of ligand, and such cells showed significant increases in IL-2 gene expression upon exposure to ligand. Thus, this initial study demonstrates that MESA may be harnessed to effectively rewire cellular input-output behavior. This work lays the foundation for leveraging MESA receptors to boost the safety and efficacy of cancer immunotherapies as well as cell-based therapies for a range of clinical applications.

Example 3

(84) Reference is made to the Abstract entitled “Modular Receptor Engineering for Programming Cell-Based Therapies to Interface with Host Physiology,” Nichole M. Daringer, Rachel M. Dudek, and Joshua N. Leonard,” presented at the 5th ICBE—International Conference on Biomolecular Engineering, Jan. 11-14, 2015, which is incorporated by reference herein in its entirety.

(85) Title: Modular Receptor Engineering for Programming Cell-Based Therapies to Interface with Host Physiology

(86) Engineered cell-based therapies represent an emerging frontier in medicine, and the promise of this approach has recently been demonstrated through curative eradication of B cell malignancy from a growing list of patients. However, extending this success to other types of cancer and to other applications in health and medicine will require new tools enabling bioengineers to custom program the manner in which engineered cells interface with host physiology. Until recently, we lacked the ability to construct customizable cell-based “devices” that detect and respond to exclusively extracellular cues, which include many species of biological relevance including cytokines, chemokines, cell-surface antigens, and pathogens. To meet this need, we developed a Modular Extracellular Sensing Architecture. This protein engineering platform comprises a self-contained receptor and signal transduction system, wherein ligand binding at the cell surface is transduced into a change in intracellular state (i.e., induction of gene expression) without requiring involvement of native intracellular signaling mediators, and therefore without being subject to native mechanisms regulating signal transduction. Here, we report our expansion upon this platform to enable the engineering of receptors that recognize essentially any external ligand and regulate essentially any native gene. We have developed general and rapidly employable engineering approaches for incorporating modular ligand binding domains (including both single-chain variable fragments, scFv, and single domain Nanobodies) to create new functional receptors.

(87) We have also engineered MESA receptors to generate “output” via Cas9-based gene regulation systems. Cas9-based transcriptional activators and repressors are easily targeted to regulate either native genes or engineered transgenes by co-expression with single-guide RNA (sgRNA) tailored to target a specific gene or set of genes. Finally, we have demonstrated that these approaches may be combined to functionally rewire mammalian cell input-output behavior. Ultimately, this capability provides a powerful tool for experimental systems biology, and in the context of medicine, enables the rapid evaluation of multiple potential strategies for functionally rewiring multicellular networks to achieve a therapeutic objective. We will present our application of this approach to modulating immune function, toward the goal of therapeutically modulating local immune states to overcome persistent barriers to immunotherapy of cancer.

Example 4

(88) Reference is made to the Abstract and Poster entitled “Rewiring cellular input-output using modular extracellular sensors,” Kelly A. Schwarz, Nichole M. Daringer, Taylor B. Dolberg, and Joshua N. Leonard, presented at the 2016 Synthetic Biology: Engineering, Evolution & Design (SEED) Meeting, Jul. 18-21, 2016, Chicago, IL, which is incorporated by reference herein in its entirety.

(89) Title: Rewiring Cellular Input-Output to Engineer Cell-Based Therapies that Interface Robustly with Human Physiology

(90) Cell-based therapies have proven to be useful diagnostic and therapeutic tools for treating a wide variety of diseases such as arthritis, infectious diseases, and cancer. Treating cancer with genetically engineered T cells is particularly promising, and for patients with certain types of “liquid” cancers, this approach has already delivered astounding clinical benefits. However, current clinical experience indicates that the extending these benefits to other types of cancer, as well as to other diseases, will require new therapeutic capabilities. In particular, there yet exists a need for technologies enabling one to program cells to sense and respond to their environment in a defined fashion. Toward this goal, our lab has developed a platform for engineered novel protein biosensors and extracellular cues, which we have termed modular extracellular sensor architecture (MESA). MESA comprises a self-contained receptor-signal transduction system in which ligand binding induces receptor dimerization, which then releases a sequestered transcription factor to regulate expression of an “output” gene or genes.

(91) Here, we presented two expansions of the basic MESA capabilities, which make key strides towards meeting the needs identified above: 1) integrating ligand-specific MESA receptors with intracellular gene circuits to enable the cell to “process” multiparametric environmental cues, and 2) rewiring cellular input-output by coupling the sensing of biologically relevant extracellular signals to modulation of endogenous gene expression. To engineer biosensing of novel input ligands, we demonstrated that both camelid antibody analogs termed “nanobodies” and scFv antibody fragments could be incorporated as novel MESA ligand-binding domains. To achieve MESA-regulated induction of endogenous target gene expression, we integrated dCas9-based transcription factors as receptor “outputs”. Altogether, these key proof-of-principle studies identified readily generalizable strategies for rewiring cellular input-output behavior in a use-defined fashion. This work lays the foundation for leveraging MESA receptors for a range of applications in both fundamental research as well as advanced cell-based therapies.

Example 5

(92) Reference is made to Schwarz et al., “Rewiring Human Cellular Input-Output Using Modular Extracellular Sensors,” *Nat. Chem. Biol.* 2017 February; 13(2):202-209, electronically published on Dec. 12, 2016, the content of which is incorporated by reference herein in its entirety.

(93) Title: Rewiring Human Cellular Input-Output Using Modular Extracellular Sensors

(94) Abstract

(95) Engineered cell-based therapies comprise a promising emerging strategy for treating diverse diseases. Realizing this promise requires new tools for engineering cells to sense and respond to soluble extracellular factors, which provide information about both physiological state and the local environment. Here, we report such a biosensor engineering strategy, leveraging a self-contained receptor-signal transduction system termed Modular Extracellular Sensor Architecture (MESA). We developed MESA receptors enabling cells to sense vascular endothelial growth factor (VEGF), and in response, secrete interleukin 2 (IL-2). By implementing these receptors in human T cells, we created a customized function not observed in nature—an immune cell that responds to a normally immunosuppressive cue (VEGF) by producing an immunostimulatory factor (IL-2). Because this platform utilizes modular, engineerable domains for ligand binding (antibodies) and output (programmable transcription factors based upon Cas9), this approach may be readily extended to novel inputs and outputs. This generalizable approach for rewiring cellular functions could enable

both translational applications and fundamental biological research.

Introduction

(96) Mammalian cell-based therapies comprise a promising and rapidly expanding technology, with over 1300 clinical trials currently underway worldwide.^{sup.1,2} Most notably, engineered chimeric antigen receptor (CAR)-expressing T cells harness the capabilities of these immune cells to detect and destroy cancer cells, which has proven transformative for the treatment of B-cell malignancies.^{sup.3-5} While extending these successes to other types of cancers poses a range of challenges, many refinements to the basic CAR approach have now been achieved.^{sup.6} However, there exist many applications for which an engineered T cell receptor cannot be utilized to achieve the desired functionality, and thus additional sensing modalities are required.^{sup.7}

(97) Technologies for engineering customized cellular functions that respond to extracellular ligands—those which are not transported into the cytoplasm—enable cells to respond in defined ways to both the local environment and overall physiological state. Such approaches generally comprise engineering native receptors to confer responsiveness to new ligands.^{sup.8-10} or coopting native receptor/signal transduction systems to drive induction of engineered signaling cascades.^{sup.11} or gene circuits.^{sup.12} While the use of native pathways is facile and leverages the performance of components tuned through evolution, these approaches are also subject to regulation and cross-talk with other native cellular functions.

(98) By contrast, orthogonal receptor systems may couple detection of extracellular ligands to engineered functions in a more programmable fashion. For example, synNotch receptors enable orthogonal detection of surface-bound extracellular ligands.^{sup.13,14} Each synNotch receptor comprises a core Notch domain flanked by modular extracellular and intracellular domains. Binding of a surface-bound ligand triggers protease-mediated cleavage and release of an exogenous transcription factor via a mechanism that requires ligand binding-mediated mechanical forces, and thus synNotch receptors do not respond to soluble ligands. We recently developed a distinct self-contained receptor-signal transduction system, the Modular Extracellular Sensor Architecture (MESA).^{sup.15} (FIG. 1a), which is also orthogonal to native cellular pathways but is designed to enable the detection of soluble ligands. Each MESA receptor comprises two transmembrane chains—the target chain (TC) and the protease chain (PC). In this system, ligand binding-induced dimerization of receptor extracellular domains promotes intracellular trans-cleavage of the target chain by the protease chain, which releases a sequestered engineered transcription factor into the cytoplasm. While this previous work demonstrated the feasibility of the MESA mechanism, ligand recognition was limited to detection of a model small molecule analyte.

(99) In this study, we investigated whether we could leverage the MESA design to generate a general platform for rewiring cellular functions in response to physiologically relevant cues. We evaluated strategies for incorporating both highly modular ligand-binding domains as MESA inputs and readily programmable transcription regulators as MESA outputs. By exploiting the modularity of MESA design, this systematic investigation both elucidated the key factors impacting the performance of these novel receptors and ultimately validated a powerful, generalizable new technology for rewiring cellular input-output functions for therapeutic applications and as a tool for fundamental research.

Results

(100) VEGF-MESA Receptor Development. In this study, we systematically developed a generalizable strategy for engineering MESA receptors that sense physiologically relevant cues. To this end, we first investigated whether we could generate ligand recognition domains by coopting the small chain variable fragment (scFv) derived from monoclonal antibodies, which confer high affinity ligand binding and are very stable.^{sup.16} scFvs have been developed against a wide range of antigens, providing a large pool of potential “parts” for receptor engineering, and such domains have been utilized successfully in other engineered receptors such as CAR T cells.^{sup.35} and synNotch receptors.^{sup.13,14} As a target analyte, we selected vascular endothelial growth factor

(VEGF). VEGF promotes angiogenesis and is present at high levels in the tumor microenvironment in many cancers.^{sup.17} Monoclonal antibodies targeting VEGF have been developed as therapeutics.^{sup.18,19}, and several are well-characterized structurally and biophysically.^{sup.20,21}. Finally, VEGF is an attractive initial target for developing MESA receptors because this ligand is a homodimer (composed of two identical protein domains), such that any scFv that binds to VEGF could conceivably be engineered into MESA chains to confer ligand binding-induced receptor dimerization.

(101) To initially develop MESA receptors responsive to VEGF (VEGF-MESA), we utilized a general strategy to assess receptor design and functionality. We initially evaluated two different anti-VEGF scFv clones, which we termed G6 and B20, after the names of the monoclonal antibodies from which we derived these scFvs.^{sup.22} (see Methods). These antibodies bind VEGF with different affinities and orientations, although each clone binds to the same face of VEGF.^{sup.20,21} (FIG. 1a). We fused each scFv clone to the extracellular portions of a MESA receptor scaffold (these constructs are hereafter termed V1-MESA and V2-MESA, respectively), including a range of flexible linkers (Gly2Ser).^{sub.x} in between the scFv and transmembrane domain.

(102) The VEGF scFV amino acid sequence of V1-MESA is as follows:

(103) TABLE-US-00002 Signal Sequence (MAWTSLLSLLALCSGASS (SEQ ID NO: 12)); V._{sub}.L (DIQMTQSPSSLSASVGDRVTITCRASQDVSTAVAWYQQKPGKA PKLLIYSASFLYSGVPSRFSGSGSGTDFTLTISSLPEDFATYYCQQGYG NPFTFGQGTKVEIKR (SEQ ID NO: 13)); GS Linker (GGGGSGGGGSGGGGS (SEQ ID NO:14)); and V._{sub}.H

(EVQLVESGGGLVQPGGSLRLSCAASGFTISDYWIHWVRQAPGK GLEWVAGITPAGGYTTYADSVKGRFTISADTSKNTAYLQMNSLRAEDTAV YYCARFVFFLPYAMD YWGQGTLLV (SEQ ID NO:15).

(104) The VEGF scFV amino acid sequence of V2-MESA is as follows:

(105) TABLE-US-00003 Signal Sequence (MAWTSLLSLLALCSGASS (SEQ ID NO: 1); V._{sub}.L (MAWTSLLSLLALCSGASSDIQMTQSPSSLSASVGDRVTITCRAS QVIRRLAWYQQKPGKAPKLLIYAASNLASGVPSRFSGSGSGTDFTLTIS SLQPEDFATYYCQQSNTSPLTFGQGTKVEIKR (SEQ ID NO: 16)); GS Linker (GGGGSGGGGSGGGGS (SEQ ID NO: 14)); and V._{sub}.H

(EVQLVESGGGLVQPGGSLRLSCAASGFSINGSWIFWVRQAPGK GLEWVGAIWPFGGYTHYADSVKGRFTISADTSKNTAYLQMNSLRAEDTAV YYCARWGHSTSPWAMDYWGQGTLLV (SEQ ID NO: 17)).

(106) The intracellular domains comprised a Tobacco Etch Virus (TEV) protease on the PC, and the cognate cleavage sequence followed by an engineered transcription factor (tTA) on the TC, as in the initial MESA design.^{sup.15} We carried this receptor library forward for evaluation.

(107) We first investigated whether VEGF-MESA receptors bound VEGF at the cell surface. When expressed in HEK 293FT cells, both V1-MESA and V2-MESA were expressed on the cell surface so long as the extracellular linker exceeded a minimal length of approximately 26 amino acids (FIG. 1(b), FIG. 4(a)). In our initial constructs, the TCs were expressed at the cell surface to a substantially greater extent than were the PCs. Given that these two chains are identical on the extracellular face and transmembrane regions, we hypothesized that the inner membrane-proximal sequence on the PC may destabilize these proteins. To test this theory, we inserted the two membrane-proximal amino acids derived from the TC (which exhibited robust surface expression) into the same location on the PC, and as predicted, this modification conferred robust surface expression of the PC (FIG. 5). Therefore, we used such a linker for all PC constructs. All V1-MESA and V2-MESA expressed at the cell surface were capable of binding recombinant VEGF, and binding did not vary substantially with extracellular linker length beyond the minimum required to achieve cell surface expression of the MESA chains (FIG. 1c, FIG. 4). Altogether, we

discovered a subset of VEGF-MESA that was expressed on the cell surface and was capable of binding extracellular VEGF, and we carried this library forward for functional evaluation.

(108) We next tested whether exposure to VEGF induced VEGF-MESA signaling. For V1-MESA, signaling in the absence of ligand increased as extracellular linker length increased from 16 to 30 amino acids, and lengthening linkers beyond this point led to no further increases in background signaling (FIG. 1d, left). Upon addition of VEGF, V1-MESA did not exhibit any ligand-inducible signaling at any linker length. The background signaling of V2-MESA was generally constant across all linker lengths evaluated and was consistently lower than the background signaling level conferred by V1-MESA (FIG. 1d, right). Most notably, for several V2-MESA configurations, we observed significant ligand-inducible signaling, with up to a 2-fold increase in reporter output upon exposure to VEGF. This observation established the fundamental feasibility of engineering MESA receptors that sense exclusively extracellular cues.

(109) We next evaluated how modifying the inter-chain cleavage kinetics might modulate VEGF-MESA signaling. Because we developed the initial intracellular VEGF-MESA architecture in order to limit background signaling in the absence of ligand^{sup.15}, we hypothesized that altering the protease cleavage kinetics, by changing the final residue of the TEV protease cleavage sequence from methionine (M) to glycine (G) (increases $k_{sub.cat}$ by ~50%^{sup.23}), could enhance signaling. VEGF-MESA with the G cleavage sequence were indeed expressed on the surface (FIG. 6(a)) and bound recombinant VEGF (FIG. 6(b)), but neither V1 or V2-MESA bearing this cleavage sequence modification exhibited ligand-inducible signaling (FIG. 6(c)). We inferred that for these G cleavage sequence MESA, transient diffusive encounters between the TC and PC resulted in cleavage and background signaling, thereby removing the dependence on the VEGF ligand to induce signaling. These results suggest a model wherein once one introduces a new ligand-binding modality, tuning MESA cleavage kinetics is neither necessary nor especially fruitful.

(110) VEGF-MESA Receptor Implementation. Having developed a functional VEGF-MESA architecture, we next turned from protein engineering to investigate how to best implement such a receptor in order to achieve desirable performance characteristics. These could include low background signaling in the absence of ligand and large fold-induction upon ligand addition, and we hypothesized that there may exist tradeoffs between these characteristics which depend on the manner in which the receptor is implemented. To frame this investigation, we posited that during the course of receptor expression, trafficking, and ligand-binding, MESA receptors may enter a number of distinct complexes (FIG. 2a). Induction of signaling requires the formation of productive dimers, where one TC and one PC dimerize at the cell surface to mediate trans-cleavage. However, since the ectodomains of both chains are identical, VEGF-binding may also induce the formation of homotypic dimers, in which two TCs or two PCs dimerize, which would not comprise functional signaling complexes. Moreover, it is also possible for TCs and PCs to encounter one another in the ER, secretory pathway, or cell surface, all of which would contribute to background signaling and generate “dead” TCs that can subsequently act as competitive inhibitors of VEGF-induced MESA signaling. We hypothesized that each of these processes would render MESA performance dependent upon the rate(s) and ratio with which each receptor chain is expressed.

(111) To investigate these questions, we systematically varied the expression of each MESA chain in a combinatorial fashion (FIG. 2b, FIG. 7(a-d)). Signaling in the absence of ligand was quite low for all VEGF-MESA receptors, although background signaling diminished somewhat for low amounts of MESA plasmids. Most notably, for both V1-MESA and V2-MESA, we observed regimes that resulted in ligand-inducible signaling. Generally, we observed the greatest ligand-inducible signaling when TC:PC expression ratios were high, and within the signaling-competent regime, increasing TC expression levels enhanced both background and ligand-inducible signaling. Using these high performing receptor expression conditions, we next investigated how VEGF-MESA signaling varied with VEGF dose (FIG. 2c). Interestingly, the dose response curves for V1-

MESA and V2-MESA were somewhat distinct. V1-MESA exhibited a substantial response to VEGF at doses as low as 25 ng/mL (FIG. 2c, left), whereas V2-MESA required a higher VEGF dose (150 ng/mL) to exhibit a substantial response (FIG. 2c, right). Moreover, V2-MESA exhibited a large dynamic range of differential responses to VEGF dose, whereas V1-MESA signaling reached a maximum at a relatively low concentration of VEGF. Possible contributors to such differences include ligand-binding affinity (V1-MESA is predicted to bind VEGF tighter than does V2-MESA, FIG. 1a) and differences between the geometric orientations in which ligand binding occurs (FIG. 1a). For the sake of comparability, we opted to use a VEGF dose of 250 ng/mL for future analyses, the lowest concentration that resulted in maximal reporter induction for both V1- and V2-MESA. Altogether, V1-MESA and V2-MESA responded to VEGF with different sensitivities and dynamic ranges, suggesting that these properties may be tuned based upon the desired application.

(112) Since tuning MESA expression levels by varying plasmid dose may not be feasible for some applications, we also investigated a method for tuning expression by varying translation rates. To this end, we introduced synthetic elements into the 5' UTR of MESA expression constructs, termed synthetic upstream open reading frames (uORF), to vary both the rate and frequency of translation initiation.^{sup.24} We screened a small library of these elements and observed that different uORF sequences did confer differential levels of MESA surface expression (FIG. 8(a,b)). We used a uORF leading to high gene expression for the TC, and we co-transfected this plasmid at a 1:1 ratio with PC plasmids encoding a variety of uORFs. Interestingly, background signaling in the absence of ligand varied with uORF choice following the same trend with which the level of MESA surface expression varied with uORF choice (FIG. 8(c)). Moreover, multiple combinations of MESA receptors containing a variety of uORFs exhibited significant VEGF-mediated reporter induction. This result confirms that tuning the relative surface expression levels of the TC and PC can confer VEGF-inducible MESA signaling and that this tuning may be implemented via multiple strategies to achieve similar MESA performance.

(113) Cellular Rewiring via VEGF-MESAs. We next investigated whether we could modify the MESA output to regulate endogenous gene expression. To this end, we replaced tTA with a readily reprogrammable transcription factor based upon CRISPR/Cas9.^{sup.25-27} In this system, catalytically inactive Cas9 (dCas9) is genetically fused to the VP64.^{sup.28} transcriptional activation domain (dCas9-TF), and expression of dCas9-TF with a small guide RNA (sgRNA) complementary to a genomic target leads to transcriptional activation of the target gene (FIG. 3a). As a proof-of-principle, we chose to target dCas9-TF to the human interleukin 2 (IL-2) gene, for two main reasons. First, IL-2 performs diverse functions in vivo including preserving peripheral immune tolerance by stimulating regulatory T cells, although high doses of IL-2 promote immune activation via stimulation of cytotoxic T cell proliferation.^{sup.29} Thus, achieving VEGF-induced expression of IL-2 is a non-natural cellular function that could be of use for immunotherapy. Second, transcription factors based upon TALE domains can reactivate the normally silenced IL-2 locus in HEK 293FT cells.^{sup.30}, such that this output comprises a suitable test of the proposed functional rewiring.

(114) Since no previous studies had systematically evaluated sgRNAs for achieving dCas9-TF-mediated transcriptional activation of IL-2 in human cells, we first developed a library of sgRNAs targeted to a region spanning upstream and downstream of the IL-2 transcription start site (positions -800 to +250, see FIG. 9) using three different in silico sgRNA design tools.^{sup.31-33} We co-expressed dCas9-TF with 15 different sgRNAs, either individually or in combination, in HEK 293FT cells, and we evaluated the induction of IL-2 mRNA expression by qPCR (FIG. 3a). Although many sgRNAs conferred minimal impacts on transcription, sgRNA 7 and 17 each individually resulted in significant induction of IL-2 transcription, as did several combinations of sgRNAs (5-10 fold). We carried the most promising sgRNAs (7, 17, 4+9+10+11) forward for further investigation.

(115) We next investigated whether recently reported methods for enhancing dCas9-TF-mediated transcription could be harnessed to enhance transcription from the endogenous IL-2 gene. To do so, we used the SAM system^{sup.34}, in which the sgRNAs were modified to contain three binding loops derived from the bacteriophage MS2 coat-protein, and two transcriptional activators (p65 and HSF1) were fused to the MS2 protein (FIG. 3b). Targeting IL-2 via SAM+dCas9-TF substantially enhanced transcription (10-15 fold) versus dCas9-TF alone (FIG. 3b), and thus we next evaluated SAM in the context of MESA.

(116) Combining the functional components identified above, we next evaluated whether VEGF-MESA output could be redirected to induce transcription of endogenous IL-2 (FIG. 3c). First, we replaced the tTA domain of the TC with dCas9-TF (V1- or V2-MESA-dCas9-TF), and we confirmed that these constructs were expressed at the cell surface (FIG. 10). To evaluate the functional performance of these receptors, we transfected V1-MESA-dCas9-TF or V2-MESA-dCas9-TF along with sgRNA 7 and SAM components into HEK 293FT cells, exposed cells to VEGF, and quantified IL-2 secretion (FIG. 3(e), FIG. 11). Monitoring IL-2 protein expression provides a robust, time-integrated metric of MESA output by circumventing potential artifacts associated with mRNA expression dynamics. We observed significant VEGF-induced IL-2 secretion for cells expressing either V1- or V2-MESA-dCas9-TF, with the V1 receptor conferring a somewhat higher overall response (~20 pg/mL IL-2 protein) and a better fold induction (5.2 fold) compared to the V2 receptor. In the presence of VEGF, both V1- and V2-MESA drove secretion of IL-2 at levels similar to that achieved by transient transfection of soluble dCas9-TF along with sgRNA 7 (FIG. 3d). Although sgRNA 17 drove high levels of IL-2 mRNA expression when co-expressed with the SAM components, transfecting sgRNA 17 in combination with V1- or V2-MESA-dCas9-TF did not result in any VEGF-induced IL-2 secretion (FIG. 12(a)). We do not yet have an explanation for this result, which was repeatable across multiple experiments. Further increasing the TC:PC ratio did not result in inducible signaling for either receptor (FIG. 12(b)). Thus, the VEGF-MESA platform successfully conferred a novel, programmed functionality in HEK 293FT cells.

(117) Finally, we investigated whether our MESA reprogramming strategy could be extended to rewire a distinct, physiologically relevant cell type. To this end, we chose to evaluate the Jurkat human T cell line, which is capable of secreting IL-2 but does not do so in response to VEGF^{sup.15}. V1- and V2-MESA-dCas9-TF were expressed on the surface of Jurkat cells (FIG. 10). We then transfected Jurkats with MESA chains, at the same TC:PC ratio used in HEK 293FT cells. We observed substantial VEGF-inducible secretion of IL-2 for both V1 and V2 receptors, and we did not observe IL-2 secretion in the absence of the MESA components (FIG. 3(g), FIG. 11). Furthermore, we confirmed that in the presence of VEGF, IL-2 accumulated only in transfected cells (FIG. 13). As we observed in HEK 293FT cells, the V1 receptor again outperformed the V2 receptor in terms of both level of IL-2 secretion and fold-induction, and when induced, each receptor drove secretion of IL-2 at levels comparable to that conferred by co-transfection of soluble dCas9-TF with sgRNA 7 (FIG. 3f). Notably, this observation represents the first functional rewiring of human immune cells to secrete an immune-potentiating molecule in response to an immunosuppressive cue, which is a response not observed in nature. Moreover, although we observed a lower IL-2 production level for Jurkat cells than for HEK 293FT cells, transfection efficiency of Jurkat cells was also substantially lower (~5% for Jurkat cells vs. 80% for HEK 293FT cells), and only the transfected Jurkats exhibited IL-2 accumulation (FIG. 13). Taking this factor into account, we estimated the VEGF-induced production rate of IL-2 to be 5 pg IL-2/10^{sup.5} transfected cells/day for HEK 293FT cells and 15 pg IL-2/10^{sup.5} transfected cells/day for Jurkat cells, such that on a per cell basis, functional rewiring may perform similarly in these two cell types. Altogether, these data demonstrate that the MESA platform comprises a generalizable synthetic biology technology for engineering cells to manifest novel, programmable, input-output functions that could be of use for both translational applications and fundamental research.

Discussion

(118) In this study, we developed a novel strategy for constructing customized cellular functions by functionally rewiring cellular input-output. Here, we consider key lessons learned in this investigation of MESA receptor design and implementation and suggest strategies for utilizing this technology in future applications.

(119) A central conclusion is that MESA's modular design and mechanism enabled the rational development of receptors recognizing novel inputs (soluble extracellular ligands). First, consider the changes required to convert the original model MESA receptors, which signal in response to rapamycin-induced dimerization of the ectodomains^{sup.36}, into MESA receptors that recognize physiologically-relevant ligands. Achieving robust cell surface expression of MESA receptors utilizing scFvs as ligand-binding domains required two key modifications: (i) inserting additional amino acids into the intracellular membrane-proximal portion of the PC and (ii) extending the nonstructured extracellular linker between the transmembrane and scFv domains of each receptor chain. We hypothesize that both of these requirements may be interpreted as providing the geometric flexibility required to enable folding of all MESA domains. Notably, extending the extracellular linker domains beyond the minimum length required to achieve surface expression did not incur observable deficits in cell surface expression, ligand binding, or ligand-inducible signaling. This is consistent with prior observations made using the model MESA receptors, all of which confirm that, as intended, MESA signaling occurs via a mechanism by which ligand binding-induced dimerization of TC and PC mediates trans-cleavage by boosting the contact frequency between these chains, rather than by constraining these chains in a geometrically defined configuration that “causes” trans-cleavage^{sup.36}. Therefore, we anticipate that MESA receptors incorporating novel scFvs need to include an extracellular linker of “reasonable” length. Notably, the scFvs on the TC and PC must bind to non-overlapping epitopes on a single ligand molecule, whether in a homotypic fashion, as was the case in this study, or in a heterotypic fashion. Moreover, the binding of the TC and PC to the analyte must bring the two chains together in a manner that enables the intracellular domains of these receptor chains to contact one another. The fact that both V1 MESA and V2 MESA exhibited ligand-inducible signaling supports the expectation that multiple geometries may meet this requirement, since the C-termini of the V1 and V2 antibodies (which about the flexible nonstructured linkers) are separated by substantially different distances when bound to VEGF (85 and 100 angstroms separation, respectively^{sup.37}). For other scFv/ligand complexes, longer extracellular linkers may be required, and for other scFv/linker complexes in which scFvs are sufficiently separated in space when simultaneously bound to ligand, it might not be possible to accommodate both ligand binding and efficient signaling. In any event, the rules identified in this study provide relatively straightforward guidance as to how design-driven modifications may be used to efficiently evaluate novel receptor designs.

(120) The modular nature of MESA design also enabled the straightforward implementation of novel receptor output (release of Cas9-based transcription factors). Since replacing tTA with dCas9-TF resulted in functional receptors with minimal adjustments, this study did not identify specific challenges associated with substituting novel transcription factors (TF) into the intracellular domain of the TC. Interestingly, we observed a higher fold induction for VEGF-MESA based upon dCas9-TF compared to those based upon tTA. One likely explanation is that tTA/TetO-based reporters are sensitive to low amounts of tTA (i.e. small amounts of transcription factor lead to moderate levels of reporter induction), which may be especially true in the context of transfection, in which many copies of the reporter exist in each cell. Conversely, dCas9-TFs are less potent and require high levels of dCas9-TF to drive robust transcription from endogenous genes^{sup.25}. Therefore, we hypothesize that the background level of TC cleavage by the PC may be comparable in each receptor system and that differences in the sensitivities of the reporter genes to released transcription factors results in higher fold-induction for the Cas9-based system when compared to the tTA-based system. We cannot exclude the possibility that some TFs may sterically

occlude TEV-mediated cleavage of the target chain (suppressing ligand-independent signaling) or that other TFs may confer less steric hindrance than does either tTA or dCas9-TF (leading to higher ligand-independent background signaling). Overall, our observations suggest that MESA output may be diversified to accommodate at least a reasonably broad range of TF structures.

(121) The manner in which a MESA receptor is implemented (i.e., expressed) also substantially impacts functional performance, in part by impacting the balance between desirable and undesirable receptor complexes (FIG. 2a). Indeed, this statement is also true of natural receptor signaling—overexpression of natural receptors via transient or stable expression of transgenes can also lead to high background (ligand-independent) signaling. Although VEGF-MESA surface expression varied relatively little when changing either plasmid dose or the uORF used, we nonetheless observed substantial changes in MESA signaling both with and without ligand.

(122) Chromosomal integration of expression vectors may better enable tuning levels and distributions of MESA expression. Since lentiviral transgene expression varies greatly with integration site, stable expression of MESA chains from a single genomic landing pad may be a preferable strategy.^{sup.38} Notably, using the homotypic binding mechanism evaluated here, ligand binding may form PC-PC and TC-TC complexes, such that a maximum of 50% of the ligand-induced dimers are signaling competent. Utilizing a heterotypic binding mechanism may circumvent this challenge, although maximizing ligand-inducible signaling would likely still require balancing TC and PC expression. Overall, implementing MESA receptors via the approaches reported here already achieved functional performance (e.g., fold-induction) that meets or exceeds that which has previously been observed with other engineered receptor systems.^{sup.39} Further exploring strategies for tuning expression within specific cellular contexts may improve functional performance without necessitating additional protein engineering.

(123) Ultimately, rewiring cellular input-output may drive both translational applications and fundamental systems biology research. By enabling one to engineer cells that respond to a target extracellular cue via the expression of either transgenes (e.g., tTA-driven) or endogenous genes (e.g., dCas9-TF-driven), MESA could facilitate the rapid implementation and evaluation of diverse therapeutic strategies. Additionally, the MESA platform could provide unique capabilities for fundamental research, particularly in the context of multicellular networks and whole organisms, including the ability to monitor, in a spatially and temporally-resolved fashion, the presence (and perhaps concentration) of VEGF in a living animal. This information would complement that observable using existing reporter strategies, which could identify where the VEGF gene is being expressed but not where the VEGF protein accumulates in the extracellular environment.

Furthermore, the MESA platform could complement genetic tools such as knockouts/ins to enable the testing of novel hypotheses pertaining to multicellular network function. Thus, MESA comprises a powerful and generalizable technology in the mammalian synthetic biology toolbox, which expands our ability to build programmable cellular functions for myriad applications.

(124) Methods

(125) DNA construct sources and engineering. Anti-VEGF scFvs and corresponding signal sequences were designed based upon the sequences of monoclonal antibodies raised against VEGF.^{sup.40}, the sequences of which were generously shared by Germaine Fuh. Inter-chain linkers of 15 non-structured amino acids were utilized to generate scFvs with strong preference for forming monovalent binding domains rather than higher order complexes.^{sup.41} These genes were synthesized (Invitrogen) and cloned into an expression vector using previously characterized MESA parts.^{sup.15} The extracellular linker (ECL) library for V1 and V2-MESA was created using PCR, digesting the product with NheI and AgeI, and inserting it back into the original backbone. The development of the tTA-responsive YFP reporter plasmid was described previously.^{sup.15} Source plasmids for Cas9 components included lenti dCAS-VP64_Blast (Addgene #61425), lenti MS2-P65-HSF1_Hygro (Addgene #61426), lenti sgRNA(MS2)_zeo backbone (Addgene #61427).^{sup.34}, MLM3636 (Addgene #43860), pMLM3705 (Addgene #47754).^{sup.25} dCas9-

VP64 (referred to here as dCas9-TF) was cloned into the MESA TC backbone in place of tTA using PCR, and digesting the product with NotI. sgRNAs were developed using overlapped primers (FIG. 7), which were inserted into either the MLM3636 or lenti sgRNA(MS2) backbone digested with BsmBI.

(126) Cell culture and transfection. HEK 293FT cells (Life Technologies/Thermo) and Jurkat cells (ATTC® TIB-152™) were maintained at 37° C. and 5% CO₂. HEK 293FT cells were cultured in Dulbecco's modified growth medium and Jurkat cells were cultured in RPMI-1640 (Life Technologies), each supplemented with 10% FBS, 1% penicillin-streptomycin, and 2 mM L-glutamine (Life Technologies). For HEK 293FT cells, transfections were performed in 6-well plates seeded at 7.5×10⁵ cells in 2 mL media (immunohistochemistry and RNA experiments) or 24 well plates seeded at 1.5×10⁵ cells in 0.5 mL media. 6-8 h post-seeding, cells were transfected using the CaCl₂-HEPES-buffered saline (HEBS) method with a total DNA content of 1-2 µg DNA per mL of media. All experiments included a plasmid expressing DsRed or BFP as a control to assess transfection efficiency. For functional experiments, 12 h post-transfection, recombinant mouse VEGF-164 (Biolegend, carrier-free) was added to cells upon media change. For Jurkat cells, transfections were performed in 6-well plates seeded at 5×10⁵ cells in 2 mL media. These cells were transfected using Lipofectamine® LTX™ with Plus Reagent, as per manufacturer's instructions. For intracellular staining of IL-2, VEGF was added 24 h post-transfection in combination with 10 µg/mL Brefeldin-A (Life Technologies).

(127) Flow cytometry, reporter assays, and immunolabeling. Adherent cells were suspended in PBS with 2 mM EDTA (PBS-EDTA) and 5% bovine serum albumin (flow buffer). Suspension cells were washed into the same flow buffer. Approximately 1×10⁶ single, live cells from each sample were analyzed using a LSRII flow cytometer (BD Biosciences) running FACSDiva™ software. Data were compensated and analyzed using FlowJo Software (Tree Star). Live, single cells were gated based on scatter and transfected cells were gated based on the presence of a DsRed or BFP transfection control for all experiments. "Reporter activity" was calculated as described in the caption of FIG. 1. For immunolabeling on the cell surface, 1×10⁶ cells were harvested and subsequently labeled with 0.5 µg allophycocyanin(APC)-conjugated anti-Flag (anti-DDDDK) tag antibody (Abcam #ab72569) for 30 minutes at 4° C., and washed 3 times with flow buffer prior to analysis by flow cytometry. For intracellular staining of IL-2, cells were first fixed and permeabilized with 4% paraformaldehyde (Sigma) for 20 minutes at 4° C., and then cells were resuspended in PBS with 0.2% BSA and 0.5% saponin (Sigma). Immunolabeling with 5 µL of an anti-IL-2 antibody (Thermo RHCIL205) was performed as described for surface staining.

(128) Statistical analysis. For most statistical analyses, two-tailed Student's t-tests were used. This test was chosen to evaluate whether there exists a significant difference between two groups of samples, and the reported comparisons meet the two requirements of this test: (1) the values compared are expected to be derived from a normal distribution and (2) the variance of each group is expected to be comparable to that of the comparison group, since the same transfection methodologies and data collection methods were used for all samples that were compared. A p value less than or equal to 0.05 was considered statistically significant. To specifically evaluate whether two flow cytometry distributions were significantly different (i.e., FIG. 1c), a chi-squared test was applied, with a significance threshold of p<0.05.

(129) VEGF binding assays. VEGF binding assays used the VEGF biotinylated fluorokine kit (R&D Systems) as per the manufacturer's instructions. Briefly, 4×10⁶ cells were harvested, incubated with biotinylated recombinant human VEGF and avidin-FITC reagents, washed to remove excess reagents, and finally re-suspended in flow buffer prior to flow cytometry analysis.

(130) Endogenous gene expression assays. RNA was isolated using Trizol® (Life Technologies), and total RNA content was determined using a Nanodrop 2000 Spectrophotometer (Thermo Scientific). 500 ng of RNA was used for cDNA synthesis using the qScript™ cDNA synthesis kit (Quanta Biosciences). 2 µL of cDNA was mixed with 12.5 µL iQ™ SYBR® Green qPCR supermix

(BioRad Laboratories), forward and reverse primers at a final concentration of 0.05 μ M, and water to a final volume of 25 μ L. qPCR reactions were run on a CFX Real-Time PCR Detection System (BioRad Laboratories). All samples were quantified in technical triplicate using IL-2 and GAPDH primers, the C.sub.q values were averaged for each primer set, and the “no reverse transcriptase” control was subtracted from the mean. IL-2 and GAPDH standard curves were used to convert from C.sub.q values to mRNA copies, and IL-2 values were subsequently normalized by the corresponding GAPDH values.

(131) Protein secretion assays. Secreted IL-2 was quantified in supernatant using the Ready-Set-Go!IL-2 ELISA kit (eBiosciences) by following the manufacturer's instructions. Undiluted supernatant was assayed in technical triplicate and a BioTek® Synergy H1 plate reader was utilized to analyze the resulting signal. Each sample was zeroed by subtracting the absorbance at 570 nm from absorbance at 450 nm, and a standard curve was used to convert between zeroed absorbance and protein concentration.

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(133) In the foregoing description, it will be readily apparent to one skilled in the art that varying substitutions and modifications may be made to the invention disclosed herein without departing from the scope and spirit of the invention. The invention illustratively described herein suitably

may be practiced in the absence of any element or elements, limitation or limitations which is not specifically disclosed herein. The terms and expressions which have been employed are used as terms of description and not of limitation, and there is no intention that in the use of such terms and expressions of excluding any equivalents of the features shown and described or portions thereof, but it is recognized that various modifications are possible within the scope of the invention. Thus, it should be understood that although the present invention has been illustrated by specific embodiments and optional features, modification and/or variation of the concepts herein disclosed may be resorted to by those skilled in the art, and that such modifications and variations are considered to be within the scope of this invention.

(134) Citations to a number of patent and non-patent references are made herein. The cited references are incorporated by reference herein in their entireties. In the event that there is an inconsistency between a definition of a term in the specification as compared to a definition of the term in a cited reference, the term should be interpreted based on the definition in the specification.

Claims

1. An exogenous extracellular sensor, comprising a target chain (TC) and a protease chain (PC), wherein the TC comprises: a) a first scFv ligand binding domain, b) a first extracellular linker comprising 26-30 amino acids, c) a first transmembrane domain, d) a first intracellular linker comprising 2-6 amino acids, wherein the first two amino acids of the first intracellular linker are the first two amino acids of a tobacco etch virus (TEV) protease cleavage sequence, e) a protease cleavage site, and f) a functional domain comprising an RNA-binding subdomain fused to a transcription regulatory subdomain; and wherein the PC comprises: a) a second scFv ligand binding domain, b) a second extracellular linker comprising 26-30 amino acids, c) a second transmembrane domain, d) a second intracellular linker comprising 2-6 amino acids, wherein the first two amino acids of the second intracellular linker are the first two amino acids of the TEV protease cleavage sequence, and e) a protease capable of cleaving the protease cleavage site of the TC.
 2. The exogenous extracellular sensor of claim 1, wherein the ligand binding domains of the TC and PC bind to a species of biological relevance.
 3. The exogenous extracellular sensor of claim 2, wherein the species of biological relevance comprises: a growth factor, a cytokine, a chemokine, a cell-surface antigen, or a pathogen.
 4. The exogenous extracellular sensor of claim 1, wherein the RNA-binding subdomain comprises a RNA-binding domain of a Cas9 protein.
 5. The exogenous extracellular sensor of claim 1, wherein the transcription regulatory subdomain comprises a transcription activator for a cognate targeted promoter.
 6. The exogenous extracellular sensor of claim 1, wherein the transcription regulatory subdomain comprises a transcription repressor for a cognate targeted promoter.
 7. The exogenous extracellular sensor of claim 1, wherein the transcription regulatory subdomain comprises a subdomain that modulates the chromatin state for a cognate targeted promoter.
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